

Jawwy TV Viewership Data Analysis

A million viewing records reveal how users engage and where the platform can grow.

Three Datasets, One Analysis

Three datasets from the Jawwy TV platform, covering March 2017 – April 2018, each addressing a distinct analytical objective.

TASK 1

Detailed Interaction Data

1,048,575 viewing records

13 columns including duration, HD quality, and genre

GOAL

Understand viewing behaviour and playback quality

TASK 2

Daily Time Series

86 business days (Jan – Apr 2018)

Total watch hours per day

GOAL

Forecast future demand

TASK 3

Ratings Data

1,048,575 records · 11,578 active users

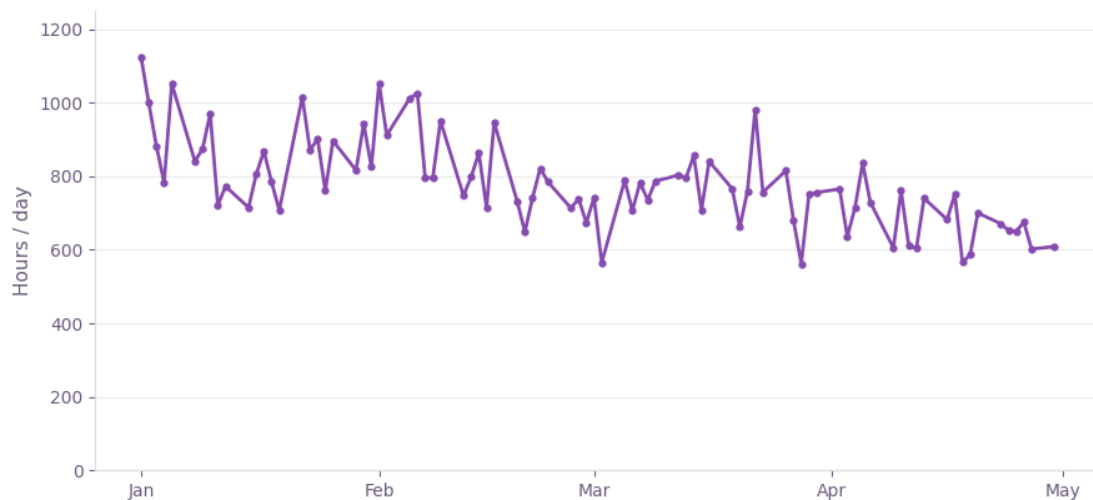
8,013 titles across 16 genres

GOAL

Build a content recommendation engine

Daily Watch Hours Show a Sustained Decline

Total daily watch hours across 86 business days, January – April 2018.



883 hrs/day

January average

first 30 days of the dataset

696 hrs/day

April average

last 30 days of the dataset

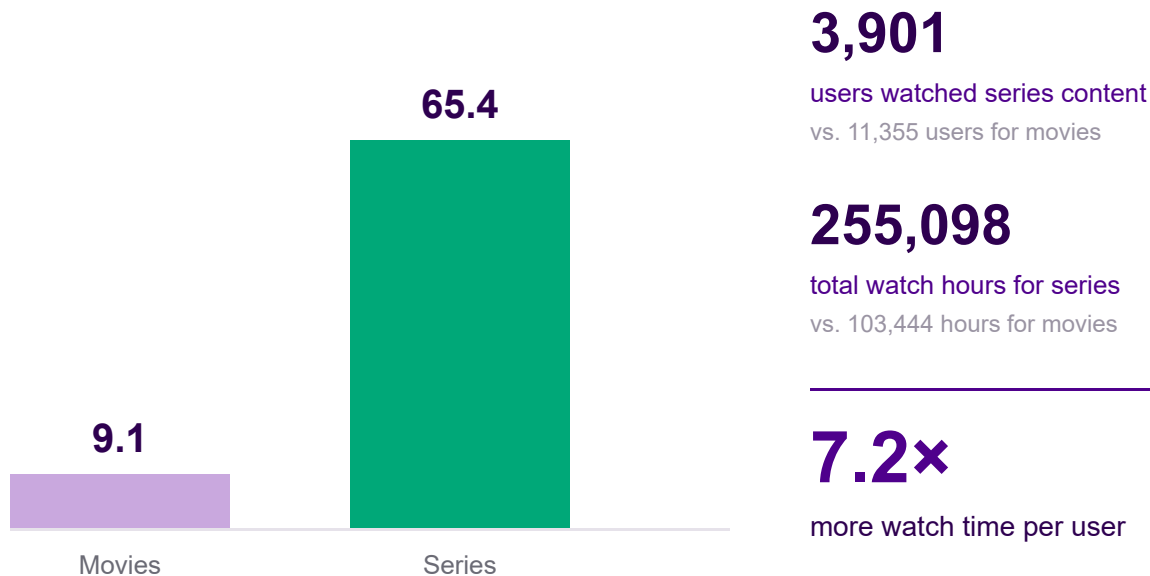
-21%

decline over four months

Series Drive Far More Engagement Than Movies

Series attract roughly one-third the viewership of movies, yet each series viewer generates substantially higher watch time.

AVERAGE WATCH HOURS PER USER



Episodic content generates substantially higher watch time per user than movies.

Top Movies Skew Toward Family and Animation

The five most-watched movies, in thousand hours of viewing time.



The Boss Baby and Moana lead the ranking.

Animation Drives Most of the Viewing

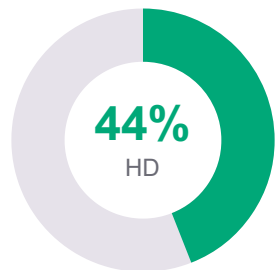
Share of all viewing records by genre.



The Playback Quality Gap

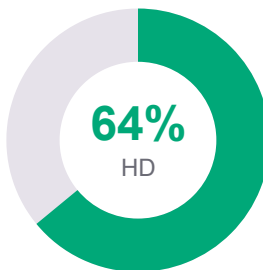
Share of viewers watching in high definition (HD) versus standard definition (SD), by program type.

Series / Episodes



56% standard definition

Movies



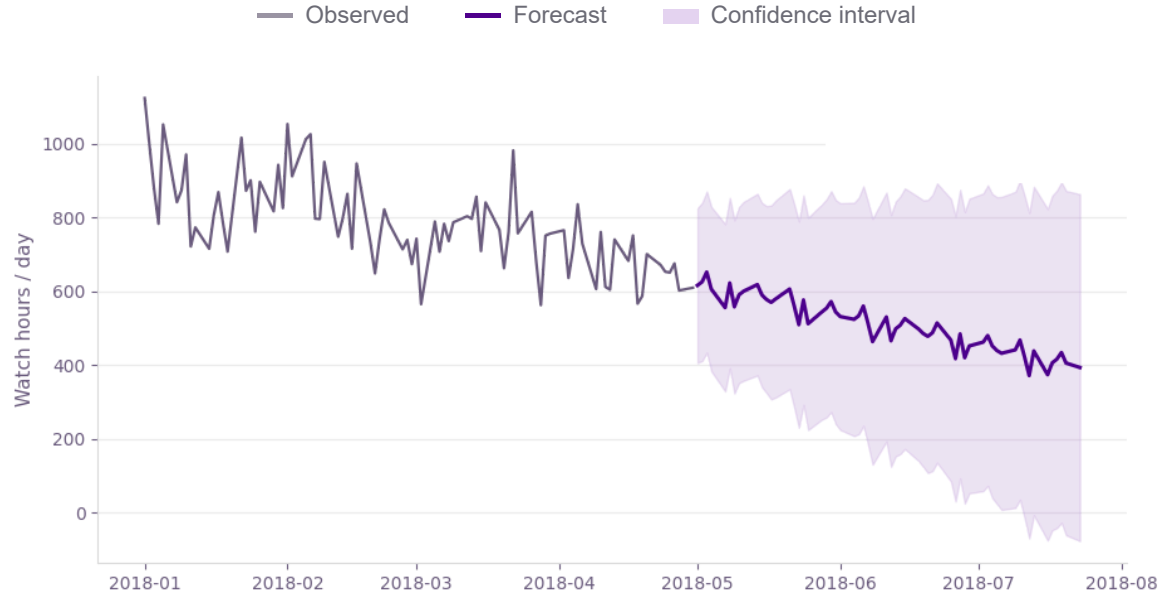
36% standard definition

WHAT IT SUGGESTS

Series lean toward standard definition far more than movies — likely a bandwidth-saving pattern during long viewing sessions.

Forecasting Future Demand

A seasonal SARIMA model projects daily watch hours two months beyond the dataset.



How the Model Was Built

From decomposition to a single AIC-selected configuration.

Time series decomposition

Separates trend from seasonal variation and noise

Selected: SARIMA(0,1,1) × (0,1,1,12)

AIC = 909.6, among the lowest scores in the search

64 SARIMA configurations

Grid search across $(p,d,q) \times (P,D,Q,12)$, evaluated by AIC

60-day forecast horizon

Confidence interval supports infrastructure and marketing planning

A Content Recommendation Engine

Collaborative filtering (K-Nearest Neighbors, cosine similarity) on the ratings of 11,578 users.

EXAMPLE

A user who watched

Moana

Family / Animation

The model identifies similar titles without manually defined rules, learning from the behaviour of users with similar preferences.

TOP 5 RECOMMENDATIONS

- 1 Trolls
- 2 Surf's Up: WaveMania
- 3 The Mermaid Princess
- 4 The Boss Baby
- 5 The Jetsons & WWE: Robo-WrestleMania!

Ranked by similarity score — how closely each title aligns with the viewing patterns of Moana's audience.

Recommended Actions

Five actions drawn directly from the findings above.

1	Deploy the recommendation engine	Launch the validated model as a “Recommended for You” row on the home screen.
2	Operationalize the forecast model	Feed SARIMA output into infrastructure and content-release planning.
3	Expand family & animation placement	Animation accounts for 38% of viewing; give it a dedicated home-screen section.
4	Investigate the HD gap in series	Determine the cause of the higher standard-definition playback rate in series.
5	Prioritize series in new investment	Series generate 7.2× the per-user watch time of movies.

Thank You

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